expending effort. In the case of massive particles interacting with the Higgs field, the interactions don't slow them down over time. Anything moving through a vacuum tends to keep doing what it's doing. In the case of massive particles, this frequently includes careening through the universe at very high (though sub-light) speeds. The main difference between massive and massless particles is that, in order to change speed, massive particles moving through a vacuum require a push, whereas massless ones travel at light speed effortlessly. In fact, massless particles can't travel at anything but light speed.

So we, who enjoy the ability to sit still once in a while, should be grateful the Higgs field did what it did, and broke the electroweak symmetry. The Higgs field not only gives particles the ability to have mass, it also determines several of the fundamental constants of nature, like the charge of the electron, or the masses of particles. The particular physical state we live in, with the Higgs field nicely situated where it is, is referred to as our "Higgs vacuum" or "vacuum state." If the Higgs field had some other value, or if the symmetry had broken in some other way, we might not be able to exist at all. We enjoy a universe in which the masses and charges of particles are perfectly set to allow them to come together in molecules, form structures, and carry out the chemical processes of life. If the field took some other value, this delicate balance might be off, potentially making these bonds impossible. We owe our entire corporeal existence to the fact that the Higgs has settled on the value it has.

And this is where things start to get a bit dicey.

Experiments like the LHC, which create extreme conditions that mimic those of the early universe, help us to see not just what the laws of physics are, but what they could be, under different circumstances. In 2012, when physicists finally were able to produce the Higgs boson in particle collisions, measuring its mass produced the final missing puzzle piece in the Standard Model of particle physics. It gave us a glimpse not only of the current value of the Higgs field, but of all the possible values it might take, given half a chance.

The good news is that the measurement of the Higgs mass is in perfect agreement with a nicely reasonable and mathematically consistent formulation of the Standard Model that has so far passed every experimental test with flying colors.